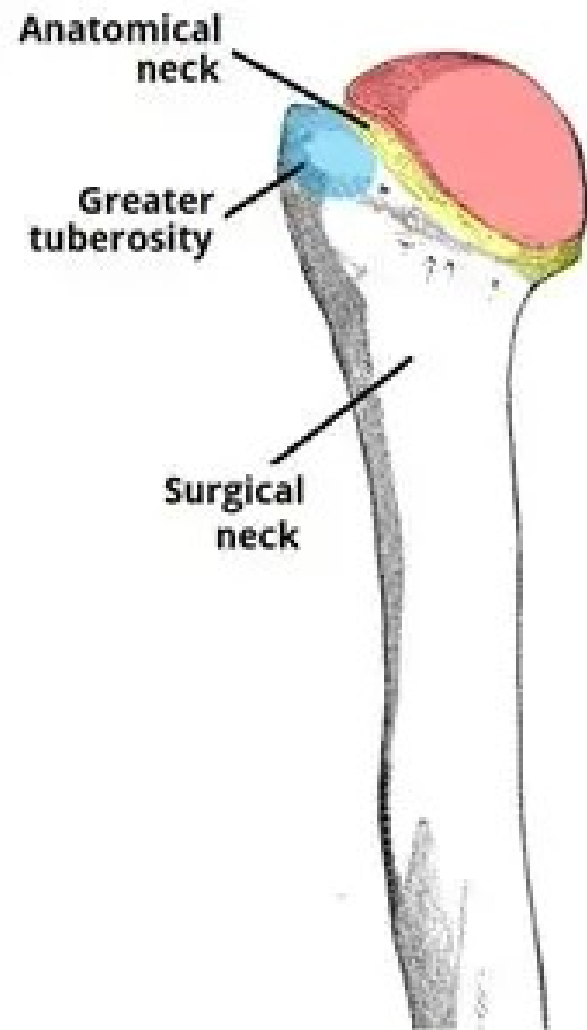


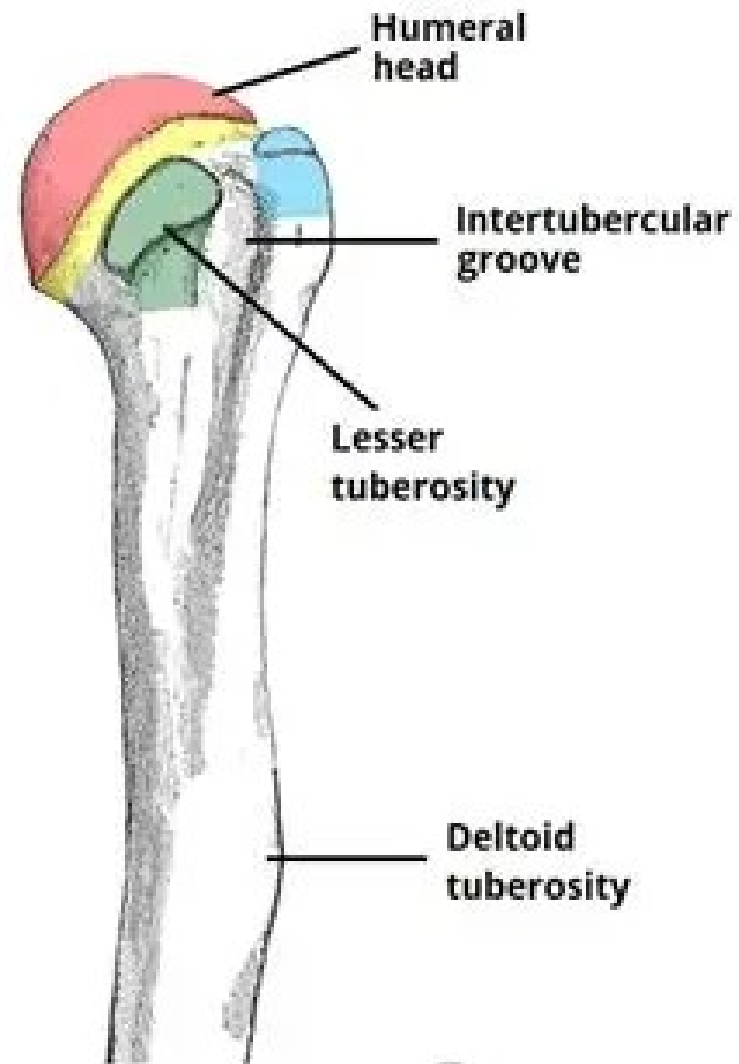
Fracture of the proximal (Neck) Humerus

Classification and Complications

- ❖ Proximal humerus fracture involving the surgical or anatomical neck.
- ❖ Neer classification is commonly used to describe the number of displaced parts.
- ❖ Important complications include nerve injury, stiffness, malunion and avascular necrosis.



(i) Posterior Face

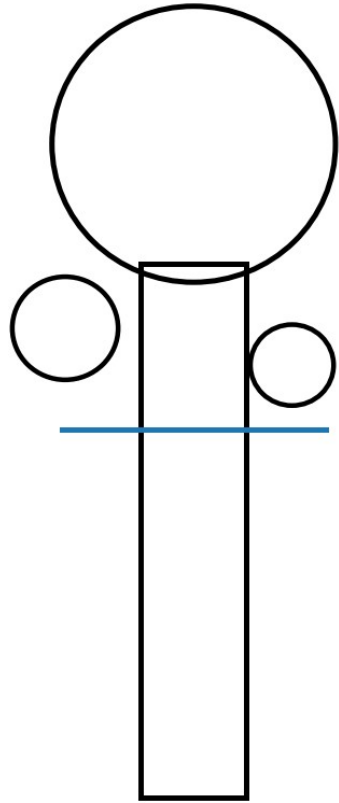


(ii) Anterior Face

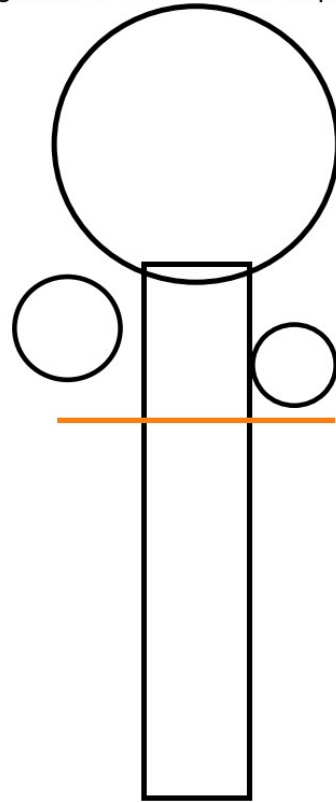
Neer Classification

Neer Classification of Proximal Humerus Fractures

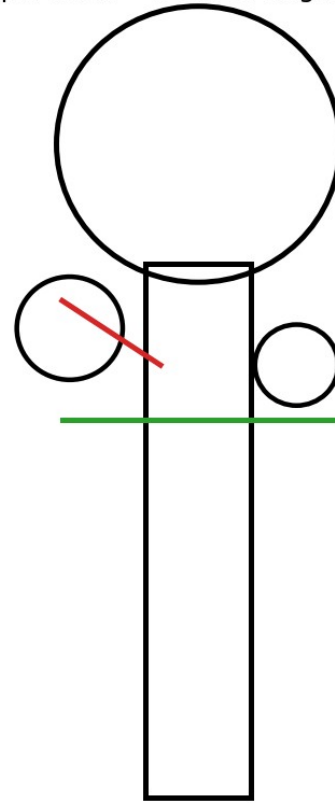
A segment is considered displaced when separation > 1 cm or angulation $> 45^\circ$



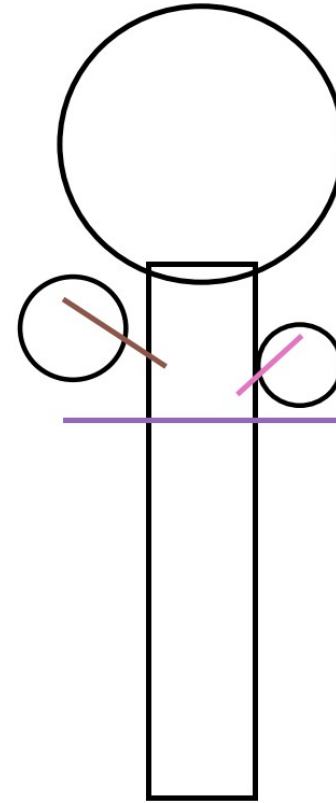
1-part



2-part



3-part



4-part

The four anatomical segments are: humeral head, greater tuberosity, lesser tuberosity and shaft.

Neer Classification of Proximal Humerus Fractures



Neer Classification — Key Points

1-part: no segment is displaced >1 cm or angulated $>45^\circ$.

2-part: one segment is displaced — commonly the surgical neck or greater tuberosity.

3-part: two segments are displaced, typically the surgical neck plus one tuberosity.

4-part: all four major segments are displaced; the humeral head may lose its normal soft-tissue attachments and treated by surgery.

Fracture-dislocations and head-split injuries are important severe patterns.

Complications

- ❖ Avascular necrosis (osteonecrosis) of the humeral head — particularly important in complex 4-part fractures and fracture-dislocations.
- ❖ Axillary nerve injury — assess sensation over the lateral shoulder and motor function when possible.
- ❖ Shoulder stiffness one of commonest complication.
- ❖ Malunion or nonunion.
- ❖ Rotator cuff dysfunction, especially with tuberosity displacement.
- ❖ Post-traumatic arthritis.
- ❖ Vascular injury is uncommon .

Clinical Assessment

- ❖ Pain, swelling, bruising and inability to move the shoulder normally.
- ❖ Look for deformity.
- ❖ Radiographs: AP and axillary view .
- ❖ CT is useful for complex 3-part/4-part fractures.

Important Take-Home Points

- ❖ Neer classification is based on the number of displaced proximal humerus segments.
- ❖ Displacement: >1 cm separation or >45° angulation.
- ❖ Four-part fractures and fracture-dislocations have greater concern for humeral-head ischemia/osteonecrosis.
- ❖ Always check the axillary nerve and distal vascular status.
- ❖ Early recognition of complications helps guide urgent referral and treatment.